Slovak University of Technology in Bratislava
Faculty of Informatics and Information Technologies

Methods of Engineering Work

Academic Year 2026/2027

Team: Matúš Čapo, Oleksii Demydchenko, Jakub Dufek, Danylo Dusheiko

Application for Applied Research (Call VV-2024)

V2X-PQC: Post-Quantum Cryptography Optimization for Secure V2X Communication in Cooperative Intelligent Transport Systems

VV-A Project identification

VV-A1-01 Project ID: APVV-24-0000

VV-A1-02 Date of submission: 1 November 2026

VV-A1-03 Project title: Post-Quantum Cryptography Optimization for Secure V2X Communication in Cooperative Intelligent Transport Systems

VV-A1-04 Project acronym: V2X-PQC

VV-A1-05 R&D specialization: Informatics (Cybersecurity)

VV-A1-06 R&D characterization: Applied research

VV-A1-07 Project start: 1 January 2027

VV-A1-08 Project end: 31 December 2028

VV-A1-09 Project annotation: V2X-PQC will evaluate post-quantum digital signatures for vehicle-to-everything communication. ECDSA is vulnerable to a sufficiently powerful quantum computer, while post-quantum alternatives introduce larger authentication payloads and different processing costs. Their suitability depends on the communication profile, traffic load and receiving device. The project will compare ECDSA P-256, ML-DSA-44 and Falcon-512 and identify where communication overhead or authentication processing limits performance.

AWSIM will provide vehicle and sensor data for defined traffic scenarios. A separate communication and authentication model will be implemented using OMNeT++ and parameterized by documented cryptographic measurements. Experiments will cover different traffic loads, including receiver workloads approaching 1,000 messages per second. The project will evaluate certificate caching and verification scheduling against a common baseline and report their benefits, limitations and security considerations. Outputs will include the experimental framework, reproducible datasets, a comparative study and a research manuscript. The results will support future V2X integration and hardware validation; production deployment and a new secure signature algorithm are not assumed outcomes.

VV-A1-10 Coordinating organization: _____

VV-A1-11 Required budget from the agency: 68,638.40 EUR

VV-A1-12 Financing from other sources: 0.00 EUR

VV-A1-13 Total project budget: 68,638.40 EUR

VV-A2 Participating organizations

Applicant

01 Name: FIIT STU Student Research group

02 Abbreviation: FSSR

03 Address: Ilkovičova 6276/2 4, 842 16 Bratislava

04 Organization ID: _____

05 Governmental branch: _____
06 Form of economy: _____
07 Contact person: _____
07 Phone: _____
07 Fax: _____
07 Email: _____
08 Statutory representative I: _____
09 Statutory representative II: _____

Cooperating organization

01 Name: Fakulta informatiky a informačných technológií Slovenskej technickej univerzity
02 Abbreviation: FIIT STU
03 Address: Ilkovičova 6276/2 4, 842 16 Bratislava
04 Organization ID: _____
05 Governmental branch: _____
06 Form of economy: _____
07 Contact person: _____
07 Phone: _____
07 Fax: _____
07 Email: _____
08 Statutory representative I: _____
09 Statutory representative II: _____

VV-A3 List of participants

Name	Titles	Role	Date of birth	Organization ID	Total hours	Hours per year
Jakub Dufek		Cryptography Researcher			600	300
Matúš Čapo		Project Coordinator			1,000	500
Danylo Dusheiko		Testing and Evaluation			600	300
Oleksii Demydchenko		V2X Developer			600	300
Other						
Total					2,800	1,400

VV-A4 Principal Investigator

01 Name and surname: Matúš Čapo
02 Gender: Male
03 Phone: 0912 456 123
04 Email: xcapo@stuba.sk

VV-B Project objectives aims and outputs

01 Key words

post-quantum cryptography, V2X, ECDSA, ML-DSA, Falcon, IEEE 1609.2, AWSIM, Unity, autonomous driving, LiDAR, certificate caching, authentication latency

02 Project objectives

V2X-PQC will evaluate post-quantum digital signatures for vehicle-to-everything (V2X) communication. It will compare processing costs and communication overhead, identify bottlenecks and test improvements to inform future V2X implementations.

AWSIM is a Unity-based simulator for autonomous driving with Autoware. It generates LiDAR, camera, IMU and GNSS data and supports ROS 2 communication. The project will use vehicle and sensor data to define V2X

scenarios. A separate model implemented using OMNeT++ will represent communication and authentication queues. Cryptographic benchmarks will run in a documented computing environment.

1. Comparison of algorithms. The project will compare ECDSA P-256, ML-DSA-44 and Falcon-512 by key, signature and secured-message sizes, signing and verification times, and memory use. Security categories and encodings will be stated. SLH-DSA will be included only in the analytical size comparison.

2. Reproducible data collection. The team will document AWSIM inputs, V2X scenarios, benchmark hardware and software, workloads and certificate-processing costs. Experiments will use at least three traffic-load levels and ten independent runs per stochastic configuration, with random seeds recorded.

3. Bottleneck analysis. The evaluation will compare verification throughput, modeled channel load and packet delivery, mean and 95th-percentile authentication latency, and authenticated-delivery ratios against predefined application targets. It will account for certificate availability and cache misses and distinguish modeled radio losses from processing delays. A 100 ms authentication deadline at receiver loads up to 1,000 messages per second will be tested as a reference scenario, with application-specific targets defined before evaluation.

4. Evaluation of improvements. Certificate caching and verification scheduling will be compared with a common baseline that defines periodic certificate transmission and digest use. Parallel verification will be tested where supported, with resource limits and trade-offs recorded to derive preliminary V2X processing requirements.

03 Indicate the project TRL level

Starting level: TRL 2 (technology concept formulated). Target level: TRL 3 (experimental proof of concept).

03 Explain the indicated project TRL level

The TRL assessment concerns the proposed V2X authentication concept. At TRL 2, its integration approach and improvement hypotheses are formulated but have not been demonstrated in the project setup.

TRL 3 requires an executable proof of concept combining AWSIM scenarios, cryptographic benchmarks and the communication and processing model. Reproducible tests must demonstrate the critical functions and evaluate at least two improvements against a documented baseline.

Results will distinguish measured processing costs from modeled communication performance. Integration into representative V2X devices, laboratory validation beyond the proof of concept and field testing remain follow-up work.

03 Project outputs and their relation to TRL

Output	Evidence	Relation to TRL
Review article	State of the art and improvement hypotheses.	Supports TRL 2.
Simulation and benchmark setup	AWSIM scenarios, processing measurements and explicit communication model.	Enables TRL 3 demonstration.
Comparative study	Performance data, bottlenecks and operating limits.	Supports TRL 3 through demonstrated functions.
Evaluated improvements	Baseline comparisons and preliminary V2X processing requirements.	Tested measures contribute to TRL 3.

Completion requires evaluation of all three main schemes and reproducible configurations and analysis scripts. Experimental post-quantum message extensions will not imply production readiness or standards compliance.

04 Project outcomes and applications in practice

The applicant is the recipient or implementer of the results

The research team will use the review, simulation setup, benchmarks and datasets to select algorithms, assess operating limits and plan integration into V2X devices and hardware validation.

Another organization is the recipient or implementer of the results

Research groups, V2X developers and automotive hardware vendors may reuse the public methodology, code and data. Transfer to other devices or deployment profiles will require further measurements and security assessment.

Other user

05 Planned publicly available outputs and contributions

Stage	Planned output	Contribution	Public availability
1	Review and experiment specification	Background, hypotheses, workloads and acceptance targets.	Manuscript and specification.
2	Simulation and benchmark setup	AWSIM scenarios, benchmark configuration, message formats and model.	Code, configurations and methodology.
3	Comparative study	Processing costs and modeled communication bottlenecks.	Report, non-personal data and analysis scripts.
4	Evaluated improvements	Baseline comparisons, quantified trade-offs and preliminary V2X requirements.	Improvement report, data and analysis scripts.
5	Final evaluation and manuscript	Reproduced comparisons, consolidated findings and follow-up guidance.	Final report, documented release and manuscript for submission.

The accompanying review establishes the scientific background; experimental outputs are planned project work. Releases will include configurations, assumptions, workloads, random seeds and reproduction instructions, subject to third-party licences.

VV-C Budget of the project in EUR

Applicant budget

Budget category	2027 EUR	2028 EUR	Total EUR
01. Direct running costs	32,619.20	33,019.20	65,638.40
02. Wage and other personnel costs	21,600.00	21,600.00	43,200.00
03. Social and health insurance	7,819.20	7,819.20	15,638.40
04. Travel costs	500.00	1,500.00	2,000.00
05. Materials	1,500.00	900.00	2,400.00
06. Services	900.00	900.00	1,800.00
07. Energy water and communications	300.00	300.00	600.00
08. Popularization costs	0.00	0.00	0.00
09. Indirect costs	1,500.00	1,500.00	3,000.00
10. Total running costs	34,119.20	34,519.20	68,638.40
Total costs from APVV	34,119.20	34,519.20	68,638.40
Financing from other sources	0.00	0.00	0.00
State outside APVV	0.00	0.00	0.00
Foreign	0.00	0.00	0.00
Private	0.00	0.00	0.00
Total costs	34,119.20	34,519.20	68,638.40

Expected costs covered by APVV

02 Personnel costs 43,200.00 EUR

The Principal Investigator will work 500 hours per year at 18 EUR per hour, totaling 18,000 EUR. The Cryptography Researcher will work 300 hours per year at 16 EUR per hour, totaling 9,600 EUR. The Software/V2X Developer will work 300 hours per year at 14 EUR per hour, totaling 8,400 EUR. The Testing and Evaluation Researcher will work 300 hours per year at 12 EUR per hour, totaling 7,200 EUR.

03 Social and health insurance 15,638.40 EUR

Employer contributions are budgeted at 36.2% of gross wages: 43,200.00 EUR $\times$ 0.362 = 15,638.40 EUR. This is the estimate used in the proposal.

04 Travel costs 2,000.00 EUR

The 2027 allocation is 500 EUR for domestic research meetings and a technical workshop. The 2028 allocation is 1,500 EUR for travel and accommodation for one scientific conference or specialist workshop.

05 Materials 2,400.00 EUR

The 2027 allocation is 1,500 EUR for development peripherals, network interfaces, cables and laboratory consumables. The 2028 allocation is 900 EUR for adapters, replacement components and additional experimental materials.

06 Services 1,800.00 EUR

The allocation is 900 EUR in each year for computing resources, hosting, storage and technical support.

07 Energy water and communications 600.00 EUR

The allocation is 300 EUR per year for electricity and network communications related to testing and computation.

08 Popularization costs 0.00 EUR

No funds are requested under this category.

09 Indirect costs 3,000.00 EUR

The allocation is 1,500 EUR per year for administrative, accounting and shared laboratory support.

Summary budget

Budget category	2027 EUR	2028 EUR	Total EUR
01. Direct running costs	32,619.20	33,019.20	65,638.40
02. Wage and other personnel costs	21,600.00	21,600.00	43,200.00
03. Social and health insurance	7,819.20	7,819.20	15,638.40
04. Travel costs	500.00	1,500.00	2,000.00
05. Materials	1,500.00	900.00	2,400.00
06. Services	900.00	900.00	1,800.00
07. Energy water and communications	300.00	300.00	600.00
08. Popularization costs	0.00	0.00	0.00
09. Indirect costs	1,500.00	1,500.00	3,000.00
10. Total running costs	34,119.20	34,519.20	68,638.40
Total costs from APVV	34,119.20	34,519.20	68,638.40
Financing from other sources	0.00	0.00	0.00
State outside APVV	0.00	0.00	0.00
Foreign	0.00	0.00	0.00
Private	0.00	0.00	0.00
Total costs	34,119.20	34,519.20	68,638.40

VV-D Project schedule

Start	End	WP	Phase
1 January 2027	30 April 2027	WP1	Analysis of V2X security requirements and post-quantum signature algorithms
1 May 2027	31 October 2027	WP2	System architecture and initial AWSIM-based V2X simulation and authentication framework
1 November 2027	30 April 2028	WP3	Integration and comparative evaluation of cryptographic mechanisms in the simulated V2X environment
1 May 2028	30 September 2028	WP4	Evaluation of improvements to verification processing and communication overhead
1 October 2028	31 December 2028	WP5	Final proof-of-concept evaluation, documentation and research manuscript

The project will run for 24 months. The work packages and milestones are described in VV-F.3.1.

Project devices and computing resources

Resource	Description
Research workstations	University computers for software development, simulation, performance measurements and documentation.
GPU computing resources	Available university or rented computing resources for simulation software and demanding traffic scenarios.
Data storage devices	Storage for simulation data, test results, measurements and project files.

VV-E Declaration of honor

Statutory representative of the applicant organization

I, the undersigned _____, statutory representative, hereby bindingly declare that:

- the organization holds a valid certificate of eligibility to carry out research and development under the conditions set out in the public call, in accordance with Section § 26a of Act No. 172/2005 Coll., as amended, and is registered in the list of persons eligible to carry out research and development;

- all information contained in the proposal documentation is true;
- the organization has, or will secure, its own resources for co-financing the project*;
- the submitted project proposal has not been and is not being funded from other sources (national or foreign);
- the organization will not provide any indirect state aid within the meaning of the specific conditions of the call;
- the organization is not a tax debtor;
- the organization is not in arrears on social insurance contributions (pension, sickness, guarantee and accident insurance, unemployment insurance, reserve solidarity fund insurance) or on contributions to old- age pension savings;
- the organization is not in arrears on health insurance contributions (with any health insurance company);
- the organization is not in liquidation;
- no enforcement of a judgment is being conducted against the organization (not relevant for public administration entities);
- no bankruptcy or restructuring proceedings have been initiated against the organization (not relevant for public administration entities);
- bankruptcy has not been declared on the organization's assets (not relevant for public administration entities);
- no petition for bankruptcy against the organization has been rejected for lack of assets (not relevant for public administration entities);
- the organization has not violated the prohibition of illegal work and illegal employment under the special regulation since its entry into force (1 April 2005) and, in the case of a violation involving the illegal employment of a foreigner under Section 2(2)(c) of Act No. 82/2005 Coll. on illegal work and illegal employment, as amended, for a period of five years after the violation of this prohibition;
- the organization has settled its financial relations with the state budget;
- the organization will cooperate in providing documents proving fulfilment of the conditions for granting the subsidy under Act No. 523/2004 Coll.;
- I am aware of the obligation to provide information on research and development funded from the state budget, for the purpose of publication in accordance with applicable Slovak legislation;
- I am aware of the obligation to publish, on the central information portal, information on the technical research and development infrastructure that we build from state budget funds or European Union funds;
- I am aware that, according to the public call, APVV will publish on its website www.apvv.sk, pursuant to Section 19d(4) of Act No. 172/2005 Coll., as amended, the decision on applications recommended and not recommended for funding by APVV bodies, limited to: application number, project title, applicant;
- I am aware of the obligation to popularize the project results in order to inform the public in an understandable way about the benefits of the R&D results and outputs;
- the content of the submitted project proposal will not be funded from national sources of grant schemes other than the relevant APVV grant scheme.

As statutory representative, I am, by my signature, responsible for the correctness and truthfulness of the information provided in this electronic application.

If requested by APVV, I undertake to replace this declaration of honour with up-to-date certificates from the relevant authorities.

I am aware that if the information in this declaration is found to be untrue, the application will be excluded, the funding agreement will not be signed, or the agreement will be terminated.

* only where there is an obligation to co-finance the project

Name of the applicant organization: FIIT STU Student Research group

Name of the statutory representative (I, II): _____

Represented by (write the name legibly): _____

Signature of the statutory representative: _____

Place: _____

Date: 1.11.2026